

Buffer circuit design; Complex CMOS (Complementary Metal-Oxide-Semiconductor) gates, CMOS building blocks: adder, multiplier; data path and memory structures

1. Introduction to CMOS Buffer

A **buffer circuit** is a digital circuit used to transfer a logic signal from one circuit to another without changing its logic value.

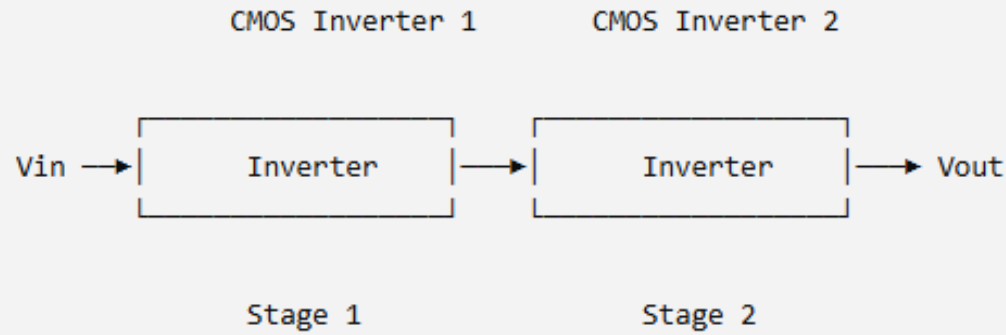
The main purposes of a CMOS buffer are:

- To increase the **driving capability** of a circuit.
- To drive a **large capacitive load**.
- To reduce the effect of load capacitance on the preceding circuit.
- To improve signal integrity and reduce propagation delay.
- To isolate one circuit stage from another.

A simple CMOS buffer can be constructed using **two CMOS inverters connected in cascade**.

2. Basic CMOS Buffer Structure

A CMOS buffer consists of two CMOS inverters:



Each inverter contains:

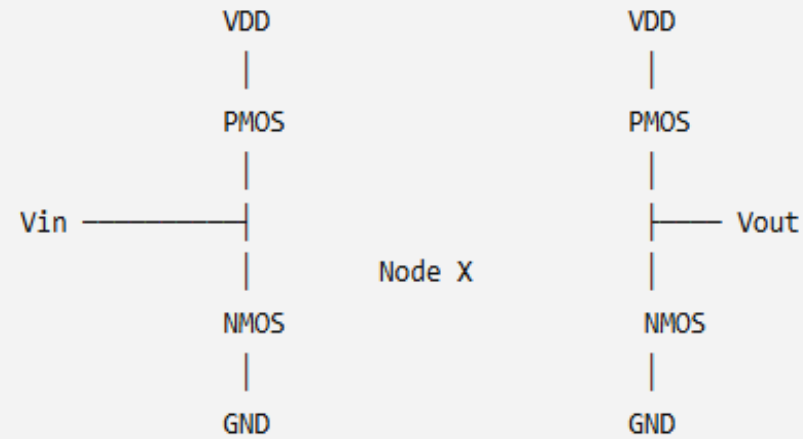
- One PMOS transistor
- One NMOS transistor

Therefore, a two-stage CMOS buffer requires:

2 PMOS + 2 NMOS = 4 transistors

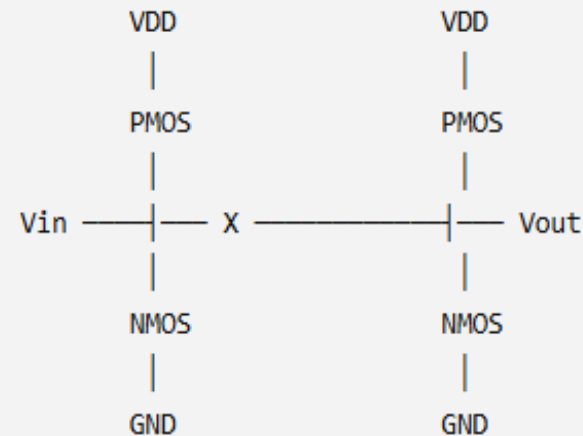
3. Circuit Diagram

A simplified CMOS buffer can be represented as:



The output of the first inverter (**Node X**) is connected to the input of the second inverter.

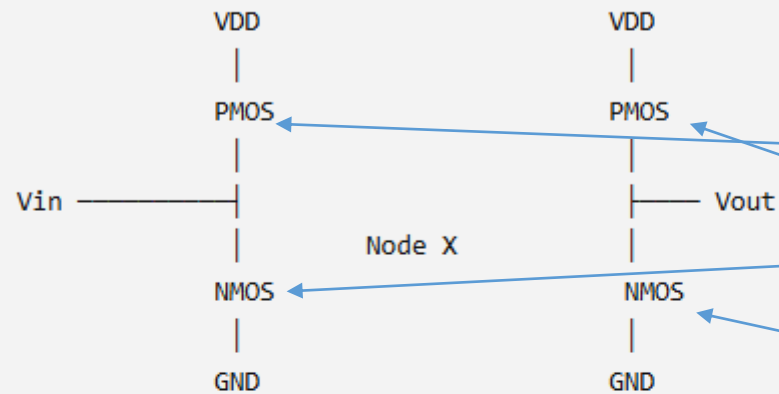
More clearly:



3. Circuit Diagram

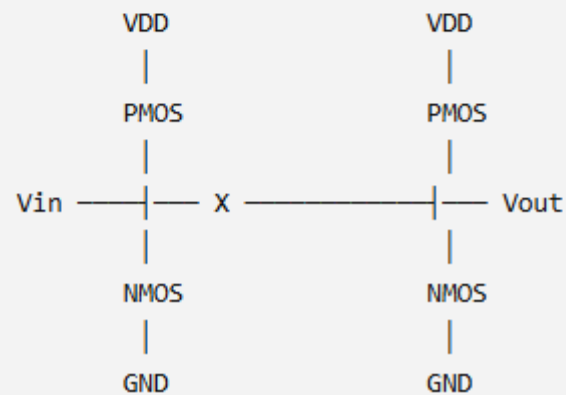
View pl

A simplified CMOS buffer can be represented as:



The output of the first inverter (**Node X**) is connected to the input of the second inverter.

More clearly:



4. Operation of CMOS Buffer

The operation can be explained using two input conditions.

Case 1: $V_{in} = 0$

For the first inverter:

- PMOS is **ON**
- NMOS is **OFF**
- Therefore, Node X = 1

For the second inverter:

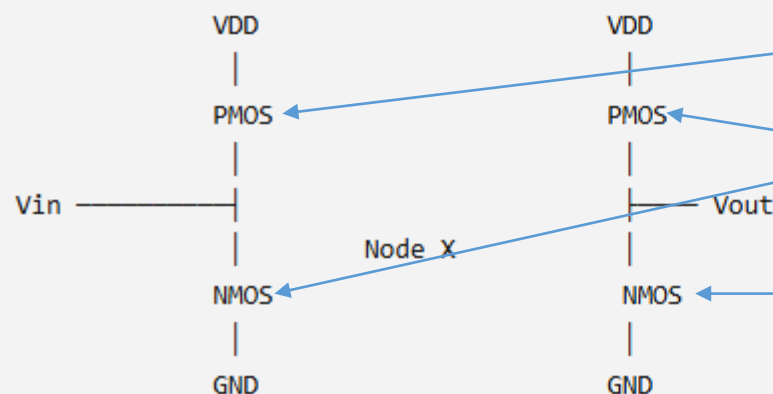
- PMOS is **OFF**
- NMOS is **ON**
- Therefore, $V_{out} = 0$

Thus,

$$V_{in} = 0 \Rightarrow V_{out} = 0$$

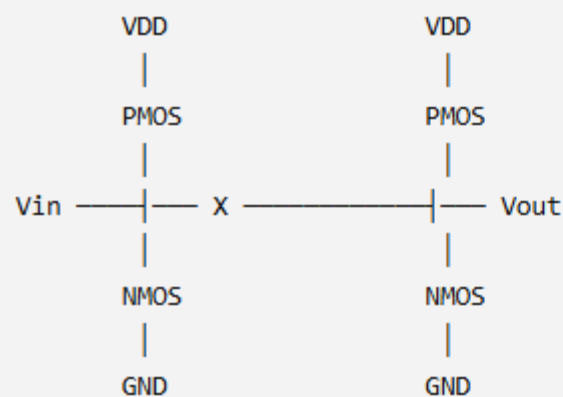
3. Circuit Diagram

A simplified CMOS buffer can be represented as:



The output of the first inverter (**Node X**) is connected to the input of the second inverter.

More clearly:



View pl

Case 2: $V_{in} = 1$

For the first inverter:

- PMOS is **OFF**
- NMOS is **ON**
- Therefore, Node X = 0

For the second inverter:

- PMOS is **ON**
- NMOS is **OFF**
- Therefore, $V_{out} = 1$

Thus,

$$V_{in} = 1 \Rightarrow V_{out} = 1$$

Therefore, the CMOS buffer performs a **non-inverting operation**.

Truth Table

V_{in}	First Inverter Output	V_{out}
0	1	0
1	0	1

Hence,

$$V_{out} = V_{in}$$

5. Transistor Sizing

One of the important aspects of buffer design is **transistor sizing**.

The second inverter must be sufficiently large to drive the required load capacitance.

A typical CMOS inverter has:

$$W_p > W_n \longrightarrow \begin{cases} W_p = \text{PMOS transistor-এর width} \\ W_n = \text{NMOS transistor-এর width} \end{cases}$$

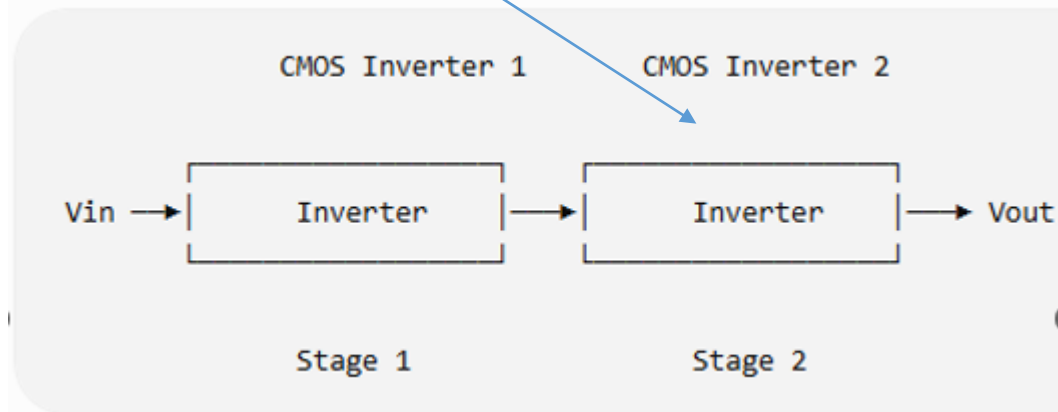
because the mobility of holes in PMOS is lower than the mobility of electrons in NMOS.

Typically,

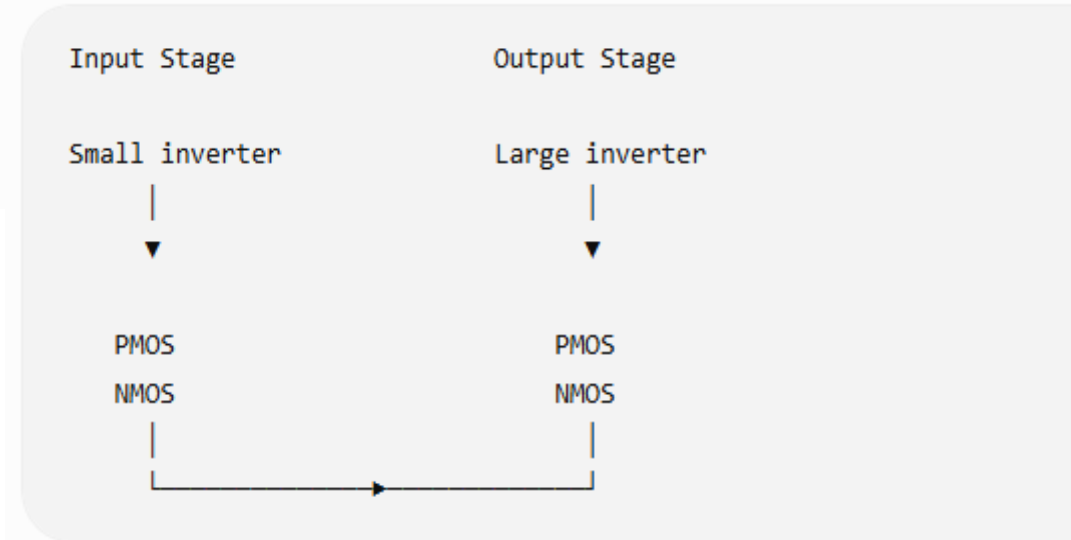
$$\frac{W_p}{W_n} \approx 2$$

although the exact ratio depends on the technology.

For a buffer, the transistors in the second stage can be made larger than those in the first stage.



For example:



This increases the current-driving capability of the output stage.

6. Driving Capability

The driving capability of a CMOS buffer depends strongly on the transistor size.

A larger transistor has a larger channel width and can provide more current.

Approximately,

$$I_D \propto \frac{W}{L}$$

where:

- W = transistor width
- L = transistor length

Therefore, increasing W increases the current-driving capability.

A buffer is particularly useful when a small logic gate needs to drive

- Long interconnects
- Large fan-out
- Large memory arrays
- Large capacitive loads
- Other high-speed digital circuits

- চিপের প্রযুক্তি যত উন্নত ও ছোট হচ্ছে, এই লম্বা তারগুলোর কারণে RC Delay (প্রতিরোধ ও ধারকত্বজনিত বিলম্ব) তত বাড়ছে। এর ফলে প্রসেসরের গতি কমে যেতে পারে এবং বিদ্যুৎ অপচয় বাড়ে।
- এই সমস্যা সমাধানের জন্য চিপ ডিজাইনাররা নির্দিষ্ট দূরত্ব পরপর Buffer (সংকেত বর্ধক) ব্যবহার করেন অথবা তামার (Copper) পরিবর্তে আরও উন্নত উপাদান যেমন কোবাল্ট বা রুথেনিয়াম ব্যবহার করছেন।

একটি একক লজিক গেটের আউটপুট সর্বোচ্চ যতগুলো লজিক গেটের ইনপুটকে নিরাপদে ড্রাইভ বা চালনা করতে পারে, তার সংখ্যা

7. Propagation Delay

The propagation delay of a CMOS buffer is mainly determined by the load capacitance and the effective resistance of the transistors.

Approximately,

$$t_p \approx 0.69 R_{eq} C_L$$

where:

- R_{eq} = equivalent transistor resistance
- C_L = load capacitance

If the load capacitance is large, the delay increases.

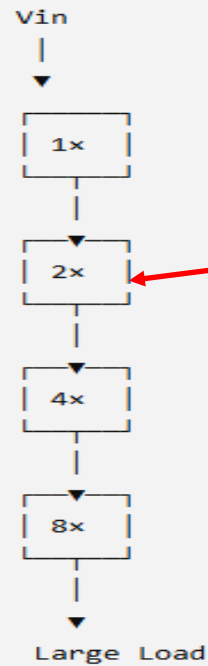
Therefore, a properly designed buffer reduces the delay by using appropriately sized output transistors.

8. Multi-Stage Buffer Design

For a very large load, using only one large inverter may not be efficient.

Instead, several progressively larger inverters can be used.

For example:



The size ratio between successive stages is selected to achieve a suitable trade-off between:

- Speed
- Power consumption
- Area

A commonly used design principle is to make each stage approximately **3–4 times larger** than the previous stage, although the optimum value depends on the technology and load.

৫. Multi-Stage Buffer (ধাপে ধাপে আকারে বড় করা)

একবারে খুব বড় ট্রানজিস্টর ব্যবহার করলে সার্কিট ধীরগতি হয়ে যায়। তাই ট্রানজিস্টরগুলোর সাইজ ধাপে ধাপে বাড়ানো হয় (যেমন: $1\times \rightarrow 2\times \rightarrow 4\times \rightarrow 8\times$).

- সাধারণত এক ধাপ থেকে পরের ধাপে ৩ থেকে ৪ গুণ বড় করা হয়।

৬. প্রধান সুবিধা ও ব্যবহার

- সুবিধা: পাওয়ার অপচয় কম, সিগন্যালে নয়েজ কমায় এবং দ্রুত গতিতে ভারী লোড চালাতে পারে।
- ব্যবহার: Clock Line, RAM/ROM Memory Bus, Processor Circuitry এবং FPGA/ASIC চিপে সিগন্যাল পৌঁছে দিতে ব্যবহৃত হয়।

9. Advantages of CMOS Buffer

The major advantages are:

1. High input impedance

একটি sensor-এর দুর্বল signal CMOS buffer-এ দিলে sensor থেকে বেশি current নিতে হয় না।

2. Low static power consumption

3. High noise immunity

4. High driving capability

একটি logic gate-এর output যদি অনেকগুলো gate-এর input-এ দিতে হয়, তখন buffer ব্যবহার করা যায়।

5. Suitable for large capacitive loads

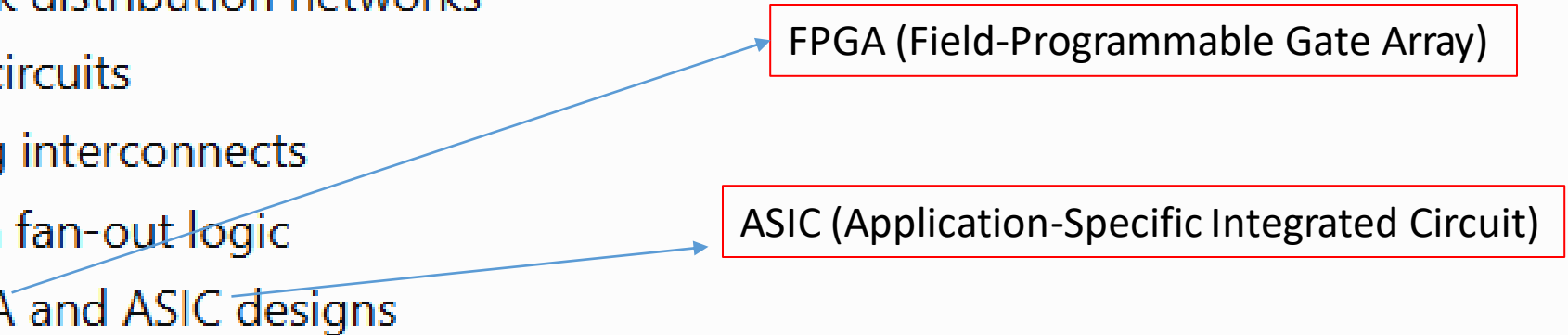
6. Improves signal propagation through long interconnects.

7. Can provide high-speed operation with proper sizing.

10. Applications

CMOS buffers are widely used in:

- Digital logic circuits
- Processor datapaths
- Memory circuits
- Clock distribution networks
- I/O circuits
- Long interconnects
- High fan-out logic
- FPGA and ASIC designs
- Address and data buses



FPGA (Field-Programmable Gate Array)

ASIC (Application-Specific Integrated Circuit)

For example, in a memory system, a buffer can be used to drive a large number of memory cells without placing excessive load on the preceding logic circuit.

11. Conclusion

A CMOS buffer is generally implemented using **two or more CMOS inverter stages** to provide a **non-inverting output with improved driving capability**. Proper transistor sizing and multi-stage design are important for reducing propagation delay and efficiently driving large capacitive loads.

$$V_{out} = V_{in}$$

Thus, CMOS buffers are essential building blocks in **high-speed digital, memory, datapath, FPGA and VLSI circuits**.

Question

A CMOS buffer consists of two inverter stages. The first inverter has an NMOS width of $1\ \mu m$ and PMOS width of $2\ \mu m$. The second inverter is designed to be **4 times larger** than the first inverter.

The load capacitance is $C_L = 20\ pF$, and the equivalent resistance of the second inverter is $R_{eq} = 50\ \Omega$.

Calculate:

(a) The PMOS and NMOS widths of the second inverter. [2 marks]

(b) The propagation delay of the buffer using

$$t_p = 0.69 R_{eq} C_L$$

[3 marks]

(c) If the load capacitance is increased to $40\ pF$, calculate the new propagation delay. [2 marks]

Total = 7 Marks

Solution

(a) Transistor sizing

First inverter:

$$W_n = 1\ \mu m$$

$$W_p = 2\ \mu m$$

Second inverter is **4 times larger**.

Therefore,

$$W_{n2} = 4 \times W_n$$

$$= 4 \times 1$$

$$\boxed{W_{n2} = 4\ \mu m}$$

Similarly,

$$W_{p2} = 4 \times W_p$$

$$= 4 \times 2$$

$$\boxed{W_{p2} = 8\ \mu m}$$

Answer:

Transistor

First Stage

Second Stage

NMOS

$1\ \mu m$

$4\ \mu m$

PMOS

$2\ \mu m$

$8\ \mu m$

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(b) Propagation Delay

Given,

$$R_{eq} = 50\ \Omega$$

$$C_L = 20\ pF$$

We know,

$$t_p = 0.69R_{eq}C_L$$

Substituting,

$$t_p = 0.69 \times 50 \times 20 \times 10^{-12}$$

$$t_p = 690 \times 10^{-12}$$

Therefore,

$$t_p = 0.69\ ns$$

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Total = 7 Marks

(c) New Propagation Delay

New load:

$$C_L = 40\ pF$$

Therefore,

$$t_p = 0.69 \times 50 \times 40 \times 10^{-12}$$

$$t_p = 1380 \times 10^{-12}$$

$$t_p = 1.38\ ns$$

Observation

আগের load ছিল $20\ pF$, delay ছিল:

$$0.69\ ns$$

Load দ্বিগুণ হয়ে $40\ pF$ হওয়ায় delay হয়েছে:

$$1.38\ ns$$

অর্থাৎ,

Load capacitance দ্বিগুণ হলে delay-ও দ্বিগুণ হয়েছে